How-To Library

Informatica

Writing data to Snowflake external tables

© Copyright Informatica LLC 2022. Informatica, Informatica Cloud, and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

This article provides instructions on how to write data to external tables in Snowflake from Cloud Data Integration.

Supported Versions

- Informatica Cloud® Data Integration Connectors

Table of Contents

Overview. 2
Create a mapping to update an external table. 2

Overview

Data in a Snowflake external table is stored in files in an external storage, while the external tables store the metadata of the data files. External tables are read-only and you can only query external tables but you cannot perform any operations directly. This article describes an example of how you can write to an external table in Snowflake using a mapping from Cloud Data Integration.

Before you write data from a source to a Snowflake external table, you need to first load the data into an underlying file in an S3, Google, or Azure storage based on where Snowflake is hosted. In Data Integration, configure a mapping to write data to the storage file using the applicable connection. You can use Amazon S3 V2 Connector, Google Cloud Storage V2 Connector, or Azure Data Lake Store Gen2 Connector based on the storage location for Snowflake.

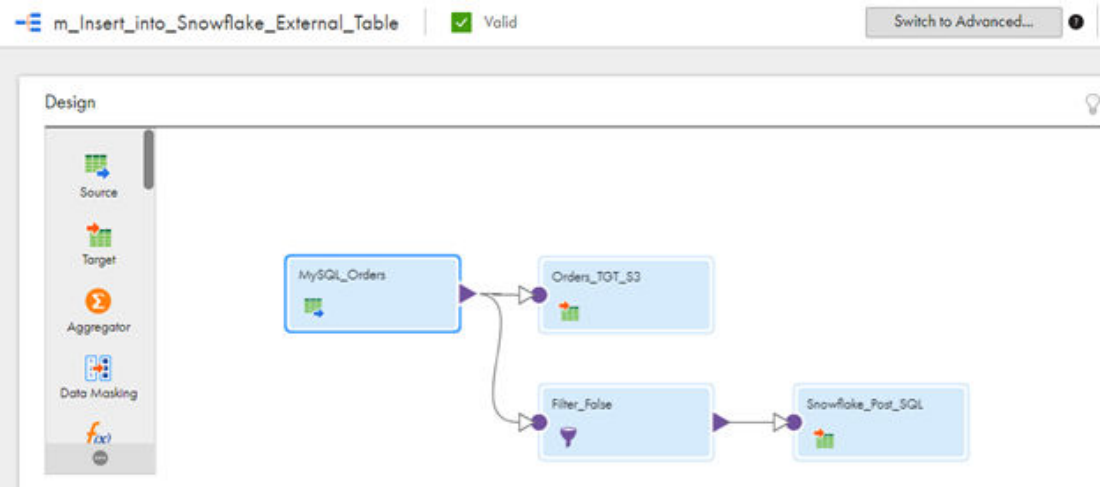
You then configure a post-SQL command in the Snowflake Data Cloud Target transformation in the mapping to update the Snowflake external tables. When you run the job, the mapping accesses the staged data files referenced in the external table definition and updates the external table metadata. You can then query the external table from Snowflake to view the updated data in the storage file.

Create a mapping to update an external table

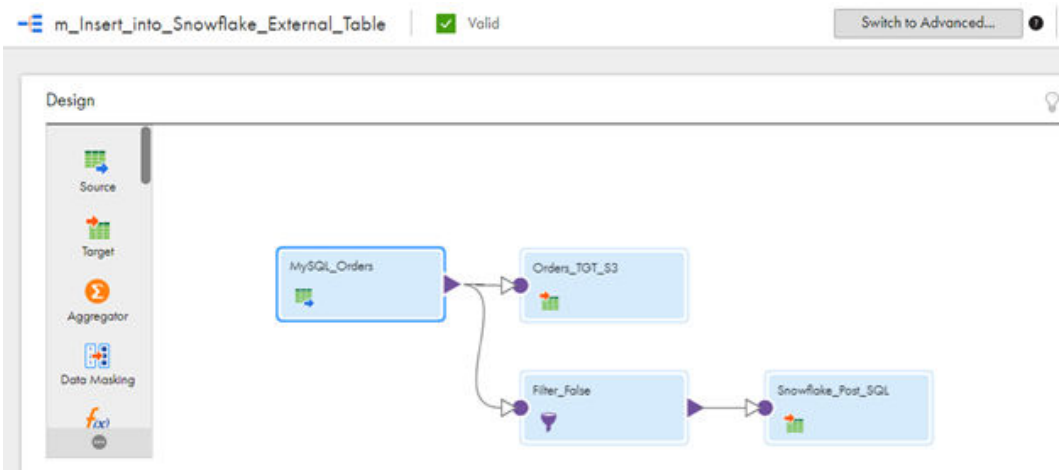
You are a sales analyst and you want to update a Snowflake external table with the product order details from a MySQL source.

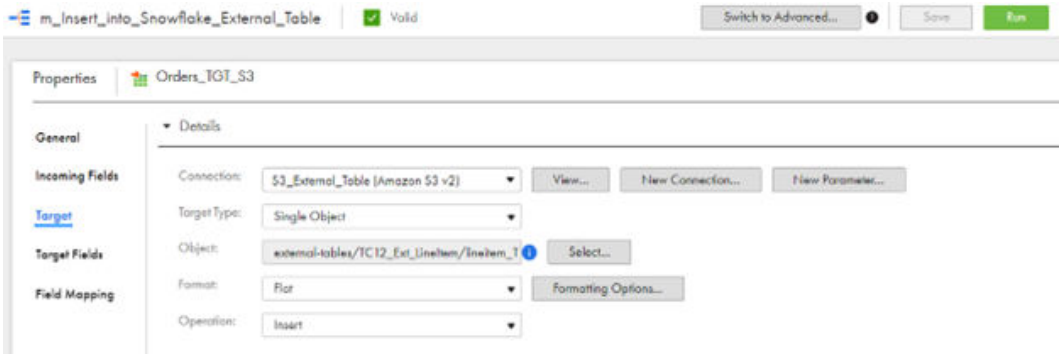
Create a mapping to read the order details from the MySQL source, stage the data in Amazon S3, and update the metadata in the external table in Snowflake.

The following image shows an example of the mapping configurations required for this use case:

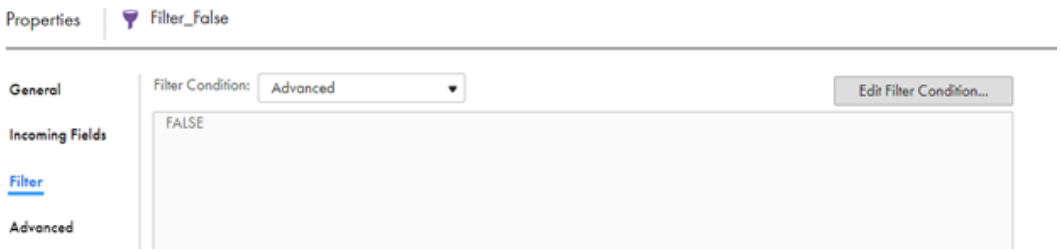


1. Configure a Source transformation to read the order details from the MySQL source.
2. Configure a Target transformation using Amazon S3 V2 connection to insert data from the MySQL source to an Amazon S3 staging file in the flat file format:

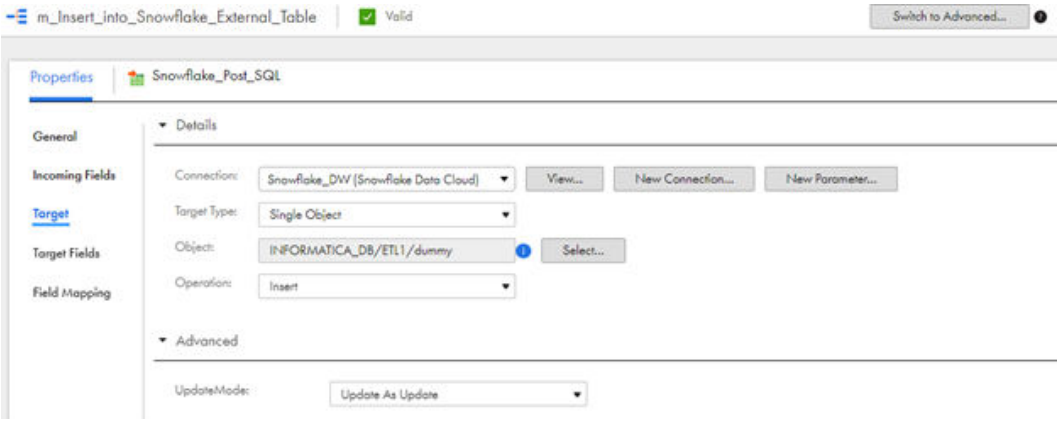




3. Configure a Filter transformation and set the advanced condition to FALSE to restrict rows to pass to the Snowflake target transformation:



4. Configure another Target transformation using Snowflake Data Cloud connection, select a placeholder empty Snowflake table as the target object, and enter a post-SQL query `ALTER EXTERNAL TABLE REFRESH` statement in the Snowflake Data Cloud advanced target properties:



For example, enter the following command in the Post-SQL field to refresh the metadata for the order details in the Snowflake external table: `ALTER EXTERNAL TABLE ORDER REFRESH`

For more information about the commands, see the Snowflake documentation: https://docs.snowflake.com/en/sql-reference/sql/alter-external-table.html

You can then query the external table from Snowflake to view the updated order details in the S3 storage.

Authors

Bharath K.R.

Dimple Rai